



Department of Computer Science and Engineering

Project Proposal of

Course Code: CSE-3636

Course Title: Artificial Intelligence Lab

AI based Chess Game using Python

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Project Title: An AI based Chess Game using Python.

Introduction: Our project is a chess game with a user-interactive interface that allows users to play against AI-generated opponents. It includes 5 distinct themes that the user can select before each match. This chess game software uses AI algorithms to analyze the current state of the game and make strategic moves based on that analysis, enabling it to play at a high level without relying on human input.

Project Background:

Chess is a classical two player strategy board game that helps us to exercise the brain, helping to improve logic, critical thinking, and problem-solving skills. It improves focus, concentration and also promotes memory and pattern recognition, as players must remember the movements of pieces and consider the potential outcomes of different moves.

Chess is now widely played as a competitive game globally, with organized tournaments and online options providing opportunities for players of all levels to compete and improve their skills. In order to participate in competitive chess, it is necessary to consistently practice and develop skills and that's why making a platform for practice is essential. Hence, we are proposing such a program so that users can practice playing chess against challenging AI opponents and improving their skills.

Objectives:

1. Our project has a user friendly interface therefore, anyone with no prior instruction can initiate playing a match by their own, guided by the interface itself.
2. It fulfills the needs of the targeted audience by giving them the chance to improve their abilities and by providing them an enjoyable gaming environment.

Complex Engineering Problem Statement (CEP):

Since we are aware that one of the requirements for this CSE-3636, Artificial Intelligence Lab course is that we must complete a project that is a Complex Engineering Problem, we have chosen a project which satisfies the criteria.

If any problem fulfills all sub criteria's of WP1 & at least two criteria's from WP2-WP7, then it is considered as a Complex Engineering Problem.

We have shown a table consisting of all the WP with their attributes including WK (Knowledge Profiles) and stated the reason why our project satisfies the criteria.

WP1 (Depth of Knowledge):

WP1 is categorized into 5 sub-attributes, which are known as Knowledge Profiles, in short WK. Game of Chess meets all the following criteria of WK's.

WK3 - Engineering Fundamental: In order to develop this project from scratch, we need to have intermediate knowledge of the Python programming language as well as the fundamentals of math, statistics, probability, and algorithms, all of which we have already covered as part of our prior course work.

WK4 - Engineering Specialist: Our project uses some exclusive algorithms to function properly.

- **Tree Generation:** The tree generator component of the chess AI generates the possible future board positions based on the possible moves that the pieces can make. At any given position, there are on average 20 possible moves that can be made in chess.
- **Evaluation Function:** This is the "intuition" of the chess AI. Given an 8x8 chess position, the evaluation function will determine whether a move is good or not by assigning it a numerical score. A negative score means that black is winning, a positive score means that white is winning. The evaluation function can take many forms depending on what approach we want to take. We can use hard coded heuristic features. We could take a model based approach (i.e. train a neural network), etc.
- **Minimax algorithm:** The minimax algorithm tries to find the optimal move from a tree of moves. Starting from the leaves of the tree which represent future states, we want to propagate back to the present root node by alternating between min () and max () (hence the name,minimax).

When it is the user's turn, he/she wants to optimize for their outcome, therefore the user wants to calculate the max () position score. When it is the opponent's (AI) turn, it wants to minimize the user's outcome, therefore it calculates the min () position score.

WK5 - Engineering Design: Our software will have a graphical user interface (GUI) design. We have broken down the design into two parts. The chessboard and the set pieces. Users can choose between four themes for the chessboard i.e., green, brown, blue, gray. The set pieces are either black or white. Users can choose the color of their pieces according to their preference.

WK6 - Engineering Practice: We are implementing this software using Python 3.9 programming language. And to develop a GUI design, we'll be developing it in Visual Studio Code.

WK8 - Literature Search: We have read & gone through some resources & previous works done on similar projects to get a better understanding. Here are the references of them -

- A guide to building a simple chess AI - <https://www.freecodecamp.org/news/simple-chess-ai-step-by-step-1d55a9266977/>
- [Algorithms Used in building Chess Game](#)
- Minimax Algorithm: <http://web.cs.ucla.edu/~rosen/161/notes/minimax.html>
- Alpha-beta Pruning: <http://web.cs.ucla.edu/~rosen/161/notes/alphabeta.html>

WP3 (Depth of Analysis Required):

To construct the AI based Chess Game software, we have to analyze some of the core algorithms and probabilistic functions in depth. One of them is Alpha beta pruning. Alpha-beta pruning is a process that can be used to greatly reduce the space of the tree search. It works by pruning the search tree as you generate it by ruling out paths that neither you nor your opponent will ever take. This is done by keeping track of two values called alpha and beta as you are searching through the tree.

Alpha is the maximum lower bound (the lowest score you are willing to accept)

Beta is the minimum upper bound (the highest score your opponent is willing to accept).

One common approach is to use Monte Carlo tree search, which uses random sampling to evaluate the various moves and select the best one based on the likelihood of winning. The AI may also use evaluation functions that assign a numerical value to the current state of the game based on factors such as piece values, material balance, and control of the center of the board, and then use this information to guide its move selection.

In addition, the AI may use machine learning algorithms such as neural networks to learn from past games and improve its performance over time.

WP7 (Interdependence):

Till now, we have thought of keeping three separate modules in the core coding of our project, which are - Pieces, rules, track. All three of them then would function together and the end product would be an AI based Chess Game.

Since, our chosen project meets the expected criteria's of CEP (WP1, WP3, WP7) we can definitely say that the Game of Chess project is indeed a Complex Engineering Problem.

Complex Problem Solving (CP)										
WP1					WP2	WP3	WP4	WP5	WP6	WP7
Depth of Knowledge					Range of conflicting Requirement	Depth of Analysis Required	Familiarity of Issues	Extensive Applicable Codes	Extend of Stakeholders involvement & conflicting	Interdependence
WK3 - Engineering Fundamental	WK4 - Engineering Specialist	WK5 - Engineering Design	WK6 - Engineering Practice	WK8 - Literature Search						
✓	✓	✓	✓	✓		✓				✓

Solution: The deliverable of our whole project is an efficient Chess practice game which serves the following outcomes:

1. A software that uses AI utilizes various statistical and probabilistic algorithms, combined with complex game tree search, to analyze the current state of the game, evaluate the opponent's moves and make strategic moves. That's why the AI can give a tough fight and challenge even experienced human players, providing a more complex and dynamic gaming experience.
2. Chess practice with other people may be constrained by scheduling conflicts and availability, but using AI makes chess practice flexible and convenient at any time.

Conclusion: The development of an AI-based chess game has the potential to provide players with both an engaging gaming environment and a platform to improve their skills. The goal of this project is to enhance the complexity and thrill of the traditional game of chess with the incorporation of cutting-edge AI technology. It has the potential to significantly advance both the area of AI and competitive chess.